

Auto Ticket Classification using Flow Designer

Technical Implementation Report & Operations Manual

EXECUTIVE SUMMARY & PROBLEM STATEMENT

The school IT helpdesk receives multiple incident requests daily from students and teachers regarding Wi-Fi issues, projector failures, password problems, and system performance. Historically, IT personnel manually reviewed each ticket to assign categories and subcategories—a process prone to human error, delays, and operational bottlenecks.

This project automates incident classification using ServiceNow Flow Designer. By evaluating keywords within incoming short descriptions, the system dynamically sets the Category and Subcategory fields, triggers targeted email responses, and streamlines ticket routing without custom code.

PROJECT OBJECTIVES

- **Automate Categorization:** Automatically assign ticket attributes at the exact moment of creation.
- **Reduce Operational Overhead:** Eliminate manual triage effort for helpdesk support agents.
- **Ensure Data Consistency:** Enforce strict Category-Subcategory dependency choices across all records.
- **Maintainability & Scalability:** Implement a robust, no-code architecture in Flow Designer that allows easy rule updates.

PHASE 1: REQUIREMENT ANALYSIS & PLANNING

1. Business Requirements

1. Parse incoming incident tickets created via self-service or backend forms.
2. Dynamically set **Category** and **Subcategory** based on keyword triggers.
3. Automatically trigger personalized email confirmation notifications to the ticket caller.
4. Standardize data fields on a dedicated table with custom auto-numbering.

2. Creation of Update Set

To preserve complete auditability and migration flexibility across development, test, and production instances, all schema and flow configurations are stored in a dedicated Update Set.

- **Navigation:** System Update Sets → Local Update Sets
- **Name:** Project Update Set
- **State:** In Progress | **Application:** Global
- **Action:** Set as Current Update Set prior to configuration.

PHASE 2: BACKEND DEVELOPMENT & CONFIGURATION

1. Table & Auto-Numbering Setup

A custom table was created to manage school IT incidents specifically formatted for workflow classification.

- **Table Label:** Incident WorkFlow | **Table Name:** u_incident_workflow
- **Auto Numbering:** Enabled (Prefix: INC, Starting Number: 500, Digits: 5).

2. Schema & Field Definitions

Field Label	Element Name	Data Type	Configuration / Reference
Number	u_number	Auto-Number	Prefix INC, Digits 5 (Read-only)
Caller	u_caller	Reference	References User Table (sys_user)
Category	u_category	Choice	[Network, Hardware, Access, Performance]
Subcategory	u_subcategory	Choice	[Wi-Fi, Projector, Forgot Password, Slow Computer]
Short Description	u_short_description	String	Free-text issue summary (Max 255 chars)
Description	u_description	String	Detailed issue description (Multi-line)
Assigned Group	u_assigned_group	Reference	References Group Table (sys_user_group)

3. Category-Subcategory Dependency Mapping

To enforce logical choices, dependency rules were mapped in the Subcategory Dictionary configuration:

- **Wi-Fi** → Dependent on Category: *Network*
- **Projector** → Dependent on Category: *Hardware*
- **Forgot Password** → Dependent on Category: *Access*
- **Slow Computer** → Dependent on Category: *Performance*

PHASE 3: FLOW DESIGNER & NOTIFICATION LOGIC

Flow Name: Auto Classify School IT Tickets | **Trigger:** Record Created (u_incident_workflow) where Category IS EMPTY.

Rule 1 (Wi-Fi / Network): IF Short Description contains "wifi" OR "network"
→ *Action:* Update Record → Category = Network, Subcategory = Wi-Fi.

Rule 2 (Projector / Hardware): ELSE IF Short Description contains "projector" OR "hardware"
→ *Action:* Update Record → Category = Hardware, Subcategory = Projector.

Rule 3 (Password / Access): ELSE IF Short Description contains "password" OR "access"
→ *Action:* Update Record → Category = Access, Subcategory = Forgot Password.

Rule 4 (Performance): ELSE IF Short Description contains "slow" OR "performance"

PHASE 4: TESTING & QUALITY ASSURANCE MATRIX

Comprehensive functional and performance tests were executed to validate workflow execution and data integrity.

Test ID	Scenario	Test Steps	Expected Result	Status
FT-01	Text Input Validation	Enter valid strings in Short Description.	Fields accept text without truncation.	PASS
FT-02	Choice Dependency	Select Category manually & check options.	Subcategory options filter dynamically.	PASS
FT-03	Flow Classification	Submit ticket with "WiFi not working".	Category auto-sets to Network / Wi-Fi.	PASS
FT-04	Email Generation	Verify email log in sys_email table.	Notification generated & queued to caller.	PASS
PT-01	Response Speed	Time flow execution upon ticket save.	Execution completes in < 3 seconds.	PASS
PT-02	Concurrent Triggers	Submit 10 tickets simultaneously.	All 10 records classified cleanly.	PASS
PT-03	Payload Load Test	Submit maximum multi-line description text.	Record updates without truncation.	PASS

PHASE 5: DEPLOYMENT & MAINTENANCE MANUAL

1. Exporting Configuration Update Set

1. Navigate to **System Update Sets** → **Local Update Sets**.
2. Open **Project Update Set**.
3. Update **State** from In Progress to Complete.
4. Click **Export to XML** under Related Links to generate the installation payload.

2. Target Instance Deployment Steps

1. In the target instance, navigate to **Retrieved Update Sets**.
2. Click **Import Update Set from XML** and upload the exported file.
3. Open the set, click **Preview Update Set**, resolve any dependencies, and click **Commit Update Set**.

CONCLUSION

The **Auto Ticket Classification using Flow Designer** implementation successfully automates ticket triage, enforces data relationships, eliminates manual overhead, and speeds up service delivery for school IT operations. The modular Flow Designer structure guarantees effortless future expansions (such as AI sentiment analysis or direct team assignment).